Experiment 06: Removable Media Security

Aim

To study and implement methods for disabling USB storage ports and write-protecting a USB flash drive using the Windows Registry Editor, the Command Prompt (DiskPart) and third-party software, and to compare how the three methods behave when used separately and together.

Objectives

1. To disable and re-enable USB storage ports by modifying Windows Registry settings.
2. To implement write protection on a USB flash drive using the Registry Editor.
3. To write-protect and remove write protection from a USB drive using DiskPart commands.
4. To study the use of third-party software for disabling USB ports and restricting write access.
5. To compare the effectiveness, ease of use, risks and reversibility of the different protection methods.

Requirements

- Windows 10/11 computer with administrator rights
- USB flash drive (SanDisk Cruzer Blade, 14 GB, FAT32, mounted as drive E:)
- Registry Editor (`regedit`) and Command Prompt with DiskPart
- USB Disk Manager (third-party utility)
- A test file (for example `test.txt`) to check copy/write access

Theory

A USB flash drive is the simplest way to move data between computers, which is exactly why it is also a common security weak point. An unrestricted port allows data to be copied out without authorisation, files on a drive to be changed or deleted by accident, and malware to enter a system from an infected drive. Two controls reduce this exposure: **port control**, which stops the operating system from using USB mass-storage devices at all, and **write protection**, which allows a drive to be read but not modified. In Windows both controls can be applied at the configuration level (Registry), at the disk level (DiskPart) or through a management utility (third-party software).

A) Windows Registry Editor

The Windows Registry is a hierarchical database in which the operating system and installed applications store their configuration. Two entries are relevant to USB storage.

USB storage driver. The key `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\USBSTOR` configures the USB mass-storage driver through its `Start` value:

- `Start = 3` – the driver loads on demand, so USB storage works (default).
- `Start = 4` – the driver is disabled, so pen drives and external disks are not recognised. USB keyboards and mice use different drivers and are not affected.

Write protection. The key `StorageDevicePolicies` under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control` holds a DWORD value `WriteProtect`. Setting it to `1` makes every attached removable drive read-only; setting it to `0` turns the policy off. This key usually does not exist by default and has to be created.

B) Command Prompt using DiskPart

DiskPart is the Windows command-line utility for managing disks, partitions and disk attributes. A disk must first be listed and selected, after which its attributes can be changed. The command `attributes disk set readonly` sets the read-only attribute on the selected disk, which blocks copying, deleting and formatting. The command `attributes disk clear readonly` removes the attribute and restores normal write access, provided no other protection is active. Because every attribute command acts on the selected disk, the correct disk must be identified from its size and details (`list disk`, `detail disk`) before any change is made; selecting the wrong one can affect important data.

C) Third-party software

Third-party utilities give a graphical front end for USB control, so access can be enabled, disabled or write-protected from a settings window instead of editing registry values or typing commands. In this experiment USB Disk Manager was used. The exact behaviour depends on the tool, and such software should be downloaded only from trusted sources and checked before installation.

D) Comparison of the methods

The three methods act at different levels. Registry settings change system configuration, DiskPart changes an attribute stored for one selected disk, and third-party software enforces restrictions according to its own settings. They are therefore not interchangeable, and removing one protection does not restore write access if another independent protection is still active.

Procedure Summary

1. Disabling USB storage through the Registry

1. Press `Win + R`, type `regedit` and open the Registry Editor with administrator rights.
2. Navigate to `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\USBSTOR`.
3. Double-click the `Start` value and change it from `3` to `4`.
4. Connect the USB drive and check This PC. The drive is not recognised.
5. Set `Start` back to `3` and reconnect the drive. The drive is recognised again.

2. Write protection through the Registry

1. Navigate to `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control` and open (or create) the key `StorageDevicePolicies`.
2. Create a DWORD (32-bit) value named `WriteProtect` and set it to `1`.
3. Reconnect the drive and try to copy a file to it. Windows reports that the disk is write-protected.
4. Change `WriteProtect` to `0`, reconnect the drive and copy the file again. The copy succeeds.

3. Write protection using DiskPart

1. Open Command Prompt as administrator and run `diskpart`.
2. Run `list disk`, then `select disk 1` for the USB drive (the 14 GB disk), and `detail disk` to confirm the device.
3. Run `attributes disk set readonly` and verify with `attributes disk` (Current Read-only State : Yes).
4. Try to copy a file to the drive. The write is rejected.
5. Run `attributes disk clear readonly`, verify that the state is No, and copy a file again. The copy succeeds. Leave DiskPart with `exit`.

4. Third-party software (USB Disk Manager)

1. Install USB Disk Manager from a trusted source and open its Preferences window.
2. Under Write Protect, choose "Turn on Write Protection", apply the setting and reconnect the drive.
3. Try to copy a file to the drive and note that the write is refused.

5. Cross-testing and comparison

The protections were then applied individually and in combination and each was removed using a different method, to see which ones reverse each other. The methods were finally compared on scope, ease of use, risks and reversibility.

Output Results

A) Registry Editor – disabling and re-enabling USB storage

In the Registry, `Start = 0x00000004`, and the USB drive is not recognised.

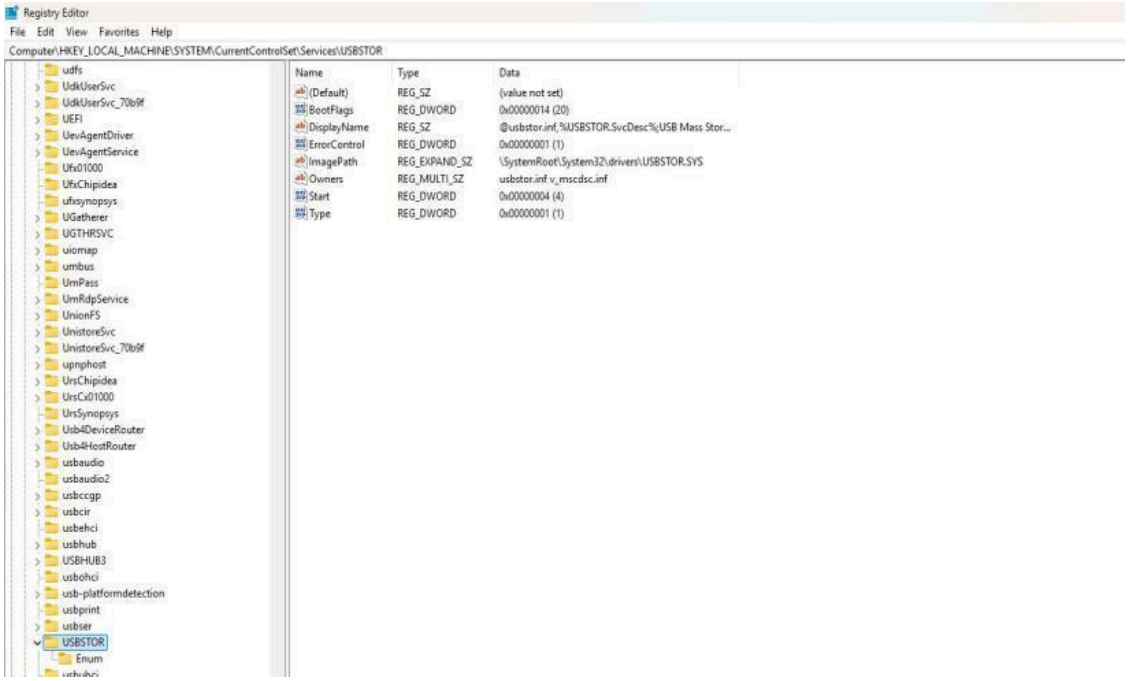


Figure 1: USBSTOR Start = 4 (USB storage disabled)

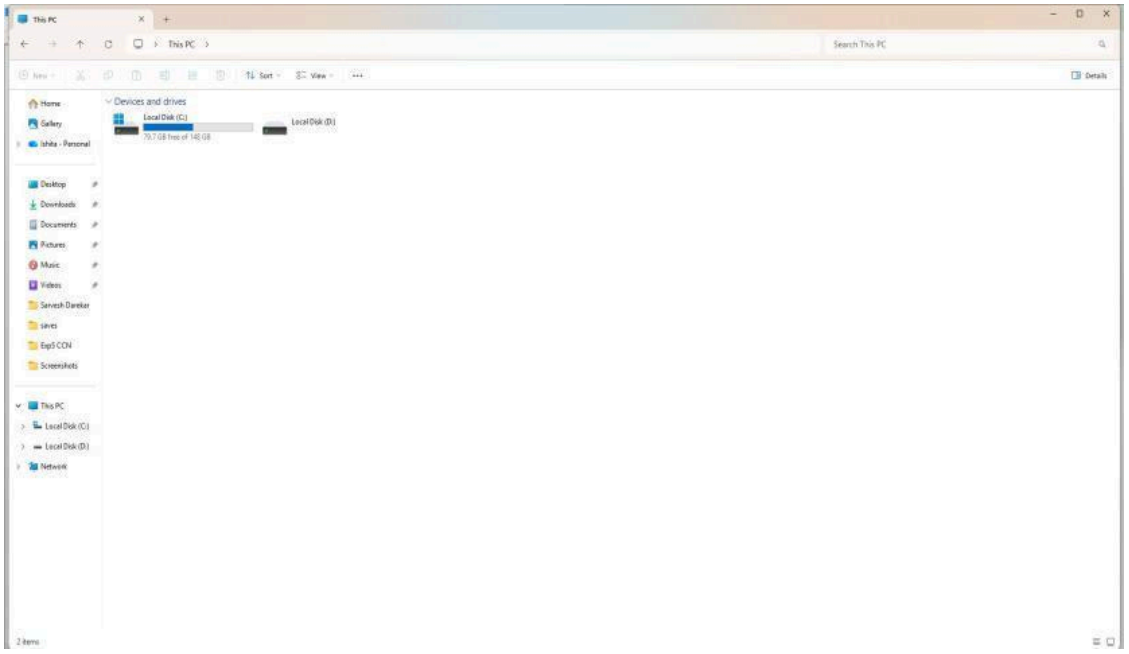


Figure 2: USB drive not recognised in This PC

After restoring `Start = 0x00000003`, the USB drive is recognised again.

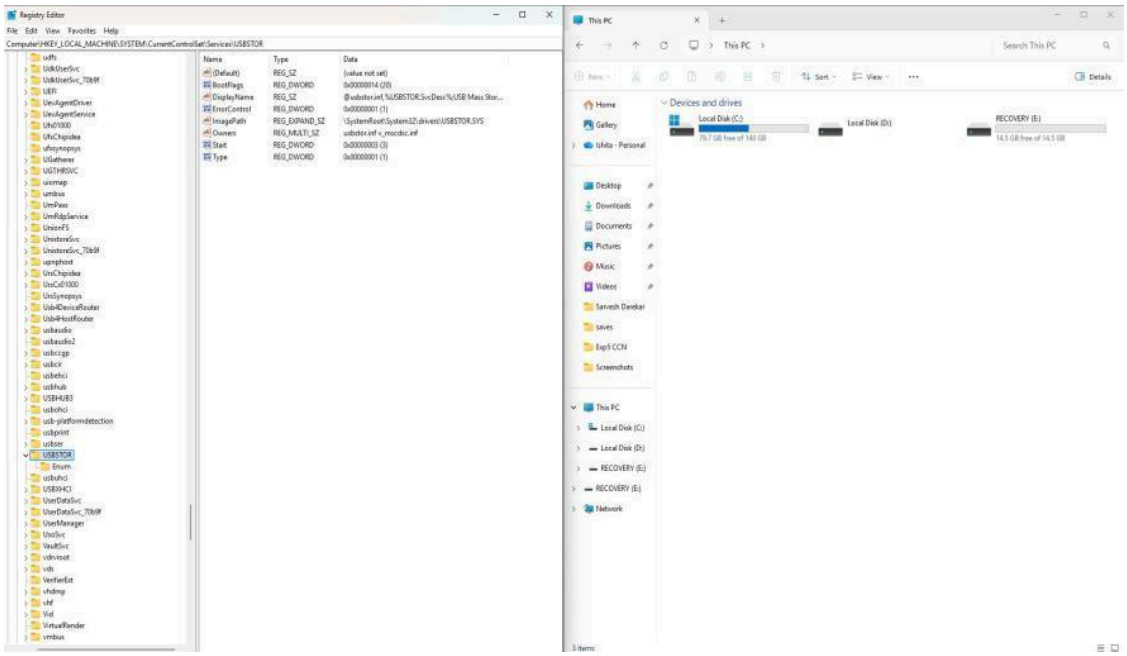


Figure 3: USBSTOR Start = 3 and drive E: visible again

B) Registry Editor – write protection

`WriteProtect = 1` (ON): files cannot be copied to the USB drive.

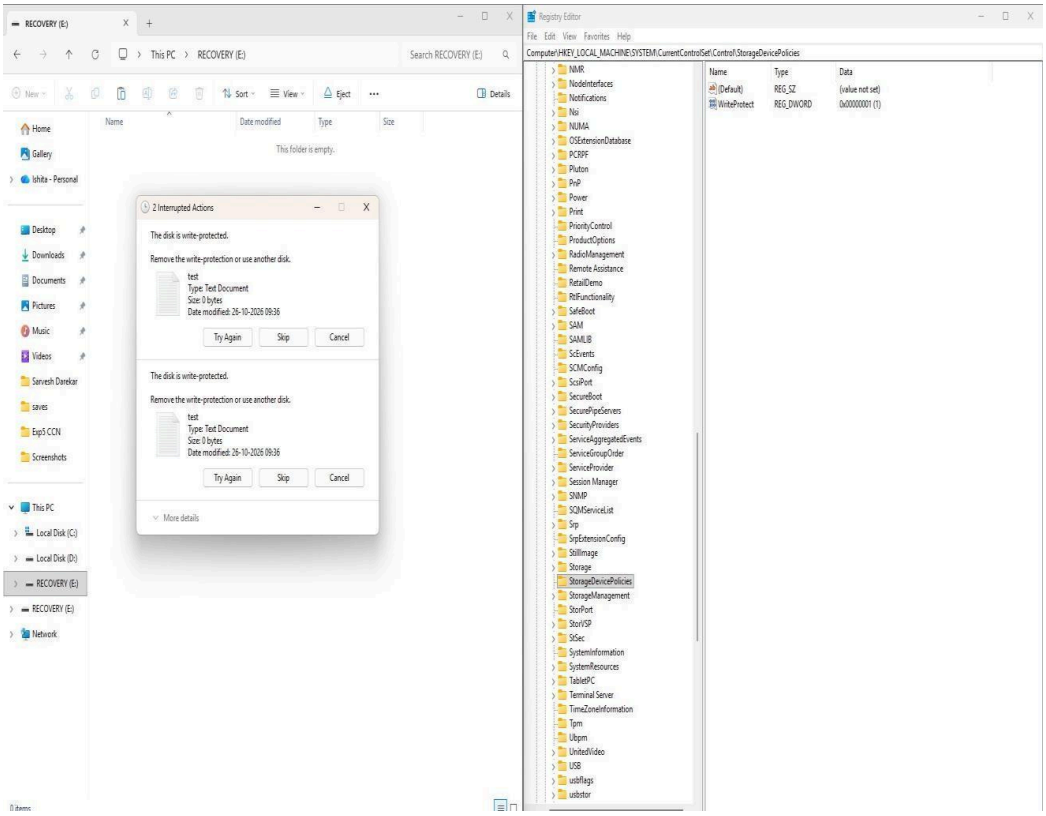


Figure 4: WriteProtect = 1, "The disk is write-protected" error

`WriteProtect = 0` (OFF): files can be copied to the USB drive.

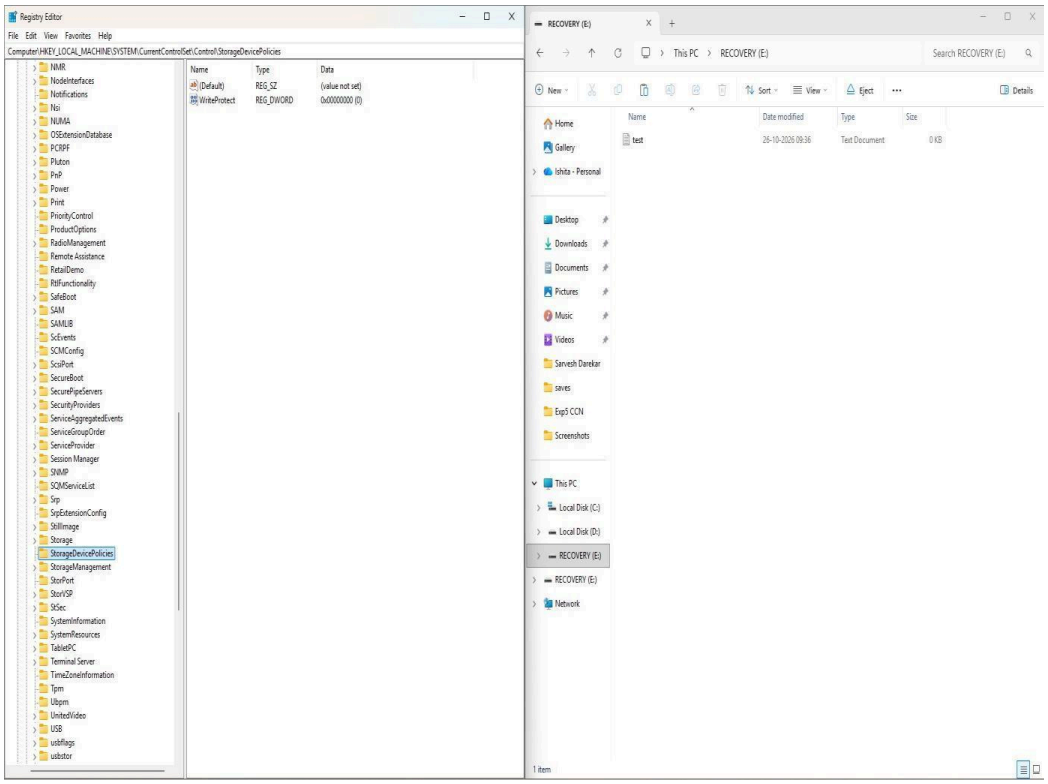


Figure 5: WriteProtect = 0, file copied successfully

C) DiskPart – command-line write protection

Disk 1 (SanDisk Cruzer Blade USB Device) was selected and made read-only. Copying a file fails with the write-protection error.

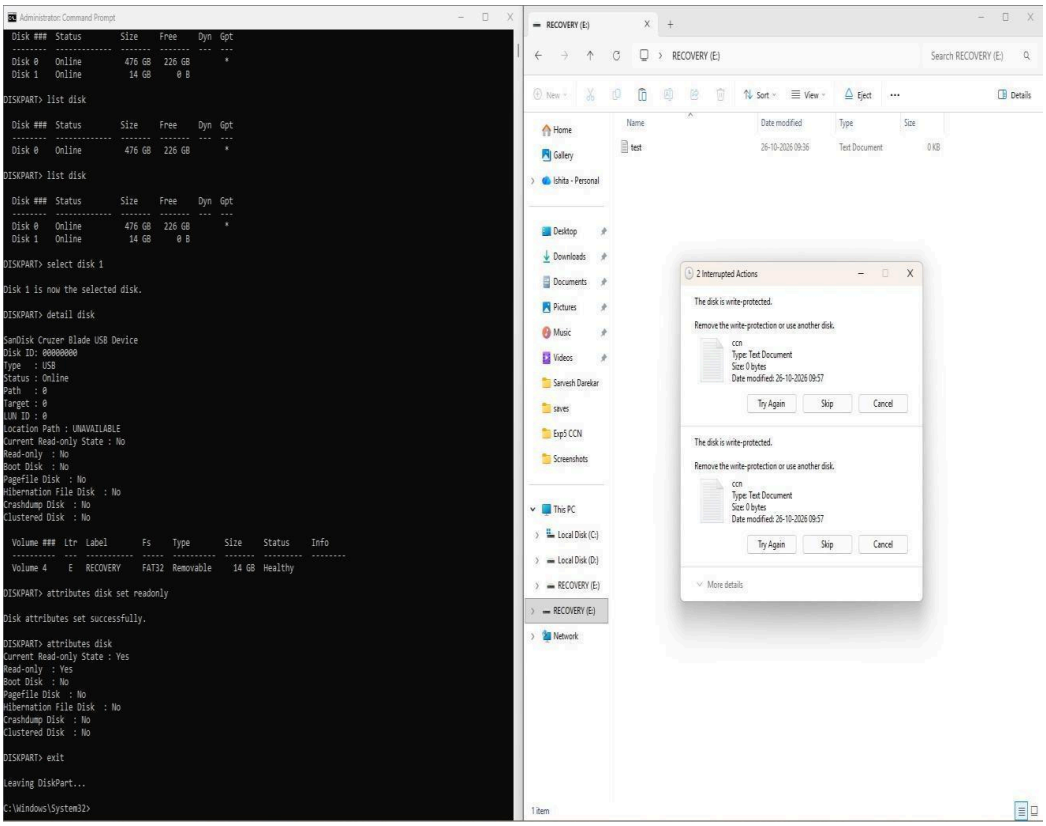


Figure 6: attributes disk set readonly (Read-only State: Yes) and failed copy

After `attributes disk clear readonly` the state returns to No and the file is copied.

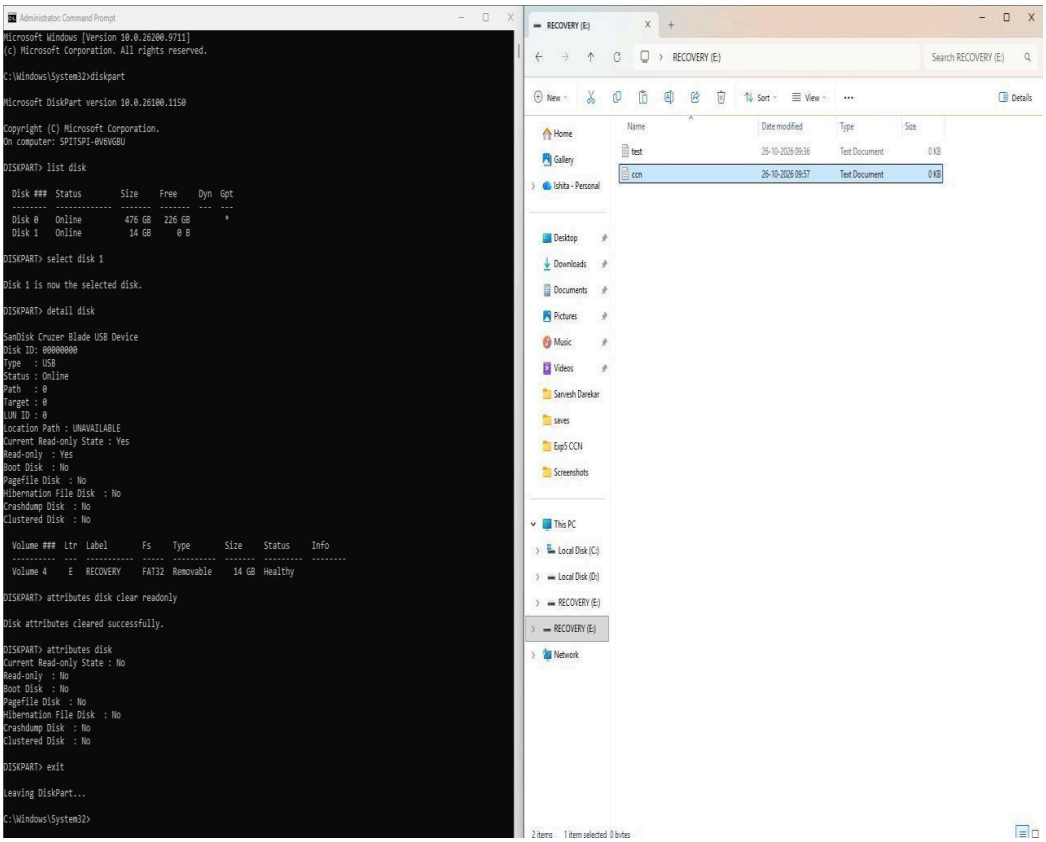


Figure 7: attributes disk clear readonly (Read-only State: No) and successful copy

D) Third-party software – USB Disk Manager

With Write Protection turned ON in USB Disk Manager, writing to the drive is refused.

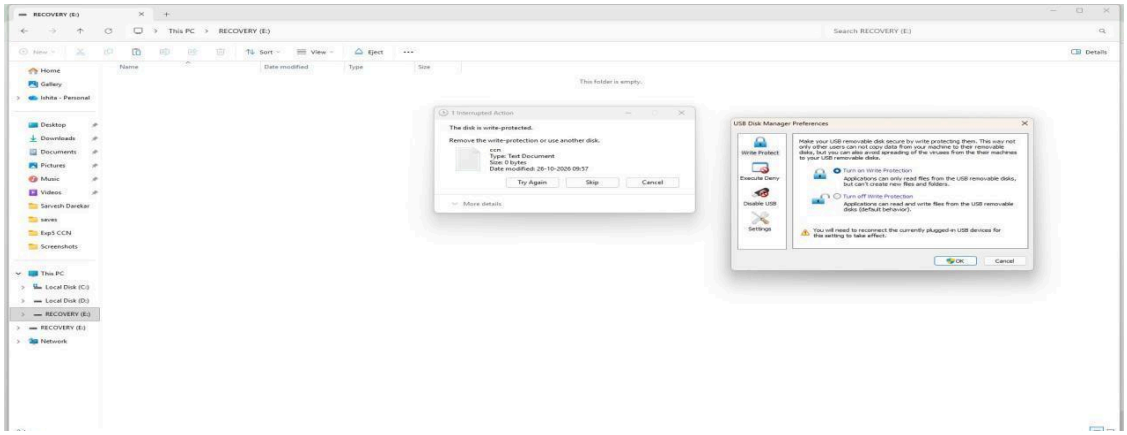


Figure 8: USB Disk Manager – Write Protection ON

Comparative Analysis 1: Cross-Reversal Test

Each protection was applied and then reversed using each of the other methods.

Protection applied	Reverse using Registry	Reverse using DiskPart	Reverse using software
Registry: `USBSTOR Start = 4`	Reversed by setting `Start = 3`; USB storage is enabled again.	Not reversed; DiskPart does not re-enable the disabled USB storage driver.	Not reversed by a write-protection toggle alone; USB storage remains disabled.
Registry: `WriteProtect = 1`	Reversed by setting `WriteProtect = 0`; writing is permitted again if no other protection is active.	Not reversed by clearing DiskPart read-only alone if Registry write protection remains active.	Not reversed by disabling software write protection alone if Registry write protection remains active.
DiskPart: `attributes disk set readonly`	Not reversed by changing Registry settings alone; the DiskPart read-only attribute remains.	Reversed by running `attributes disk clear readonly`.	Not reversed by changing software settings alone if the DiskPart attribute remains active.
Third-party software	Depends on the software setting. Registry changes will not necessarily undo software-enforced restrictions.	Depends on the software setting. Clearing DiskPart read-only does not necessarily undo software restrictions.	Reversed by disabling the corresponding protection in the software, provided no other restriction remains.

Comparison of the 3 Methods

Criterion	Registry edit	DiskPart	Third-party software (USB Disk Manager)
Affects	`USBSTOR Start = 4` affects USB mass-storage devices system-wide. `WriteProtect = 1` applies a write-protection policy.	Affects the selected disk only.	Depends on the software settings; can control USB access or write protection.
Ease of use and time taken	Moderate; requires navigating Registry Editor and changing values.	Moderate; requires administrator access and command-line instructions.	Easy; provides a graphical interface and is convenient for demonstrations.
Risks or side effects	Incorrect Registry changes can affect system functionality or block USB storage access.	Selecting the wrong disk can affect important data.	Untrusted software may introduce security risks; incorrect settings may restrict access.

Criterion	Registry edit	DiskPart	Third-party software (USB Disk Manager)
Reversed by which other methods?	Generally requires changing the relevant Registry value back. Other methods do not automatically undo it.	Requires `attributes disk clear readonly`; Registry or software changes do not necessarily remove the disk attribute.	Generally requires disabling the relevant option in the software; other methods may not remove its restrictions.
Best suited for	System-level USB storage restrictions and Registry-based write-protection policies.	Write-protecting an individual USB drive.	Quick demonstrations and convenient graphical control of USB restrictions.

Two Methods Together

Methods used: Registry write protection (`WriteProtect = 1`) and DiskPart read-only protection (`attributes disk set readonly`).

After removing the first protection	After removing the second protection
After setting `WriteProtect = 0`, the USB drive remains read-only because the DiskPart read-only attribute is still active. Copying files to the drive is still restricted.	After executing `attributes disk clear readonly`, both protections have been removed. Normal write access is restored, provided no other protection is active.

Questions and Solutions

Q1. Which method is best for (a) a shared laboratory PC, (b) protecting a single pen drive, and (c) a quick classroom demonstration? Justify each choice.

Solution:

(a) **Shared laboratory PC – Registry.** Setting `USBSTOR` to `Start = 4` stops USB mass-storage devices from being accessed through the driver on that computer, so pen drives cannot be used by any user, while keyboards and mice continue to work normally.

(b) **Single pen drive – DiskPart.** DiskPart applies the read-only attribute to one selected disk. That drive cannot be written to, but USB storage does not have to be disabled for the whole system.

(c) **Quick classroom demonstration – third-party software.** A tool such as USB Disk Manager lets the user switch protection on and off from a graphical window, without navigating the Registry or entering several commands.

Q2. Did any method reverse the effect of another? Explain why or why not.

Solution: No. The methods control different settings: the Registry Editor changes Windows configuration values, DiskPart changes the read-only attribute of a selected disk, and the third-party software applies restrictions through its own settings. In the tests, removing Registry write protection did not restore

write access while the DiskPart read-only attribute was still active, and clearing the DiskPart attribute did not re-enable USB storage while the USBSTOR driver remained disabled (see the cross-reversal table). Each active protection therefore has to be removed separately.

Q3. Can an administrator undo all three methods? What would you use if stronger protection were required?

Solution: Yes. An administrator with the necessary permissions can undo all three by restoring the relevant Registry values, clearing the DiskPart read-only attribute and disabling the corresponding restrictions in the third-party software, although other active restrictions or organisational policies may also need to be addressed. For stronger protection I would use centrally managed device-access policies, restrict permissions to authorised users and encrypt sensitive data, since these are stronger than relying on Registry settings or basic write protection alone. Hardware-enforced write protection (a physical write-protect switch) may also be used where supported.

Summary

Three methods of controlling USB storage access were implemented: Registry Editor, DiskPart and third-party software. The Registry was used to disable the USB mass-storage driver and to enable write protection, DiskPart made a selected disk read-only, and USB Disk Manager offered a quick graphical way to manage the same restrictions. File-copy tests confirmed each effect, and normal access was restored at the end of the experiment.

The tests also showed that each method works independently: removing one restriction does not remove the others. The Registry method suits restricting USB use on shared computers, DiskPart suits protecting an individual drive, and third-party software suits simple, convenient control.

Conclusion

In this experiment we studied how USB storage access can be controlled in Windows using the Registry Editor, DiskPart and third-party software. We disabled and re-enabled the USB mass-storage driver, applied write protection to removable drives and to a selected disk, and checked each result by trying to copy files.

The Registry acts at system level, DiskPart acts on one particular drive, and third-party software gives an easier graphical way to manage the restrictions. Because the protections are independent, they must be removed separately to restore normal USB access. These measures help prevent unauthorised data transfer and accidental changes to stored files, though stronger environments need centrally managed policies, encryption and access control in addition.